Lightweight SIEM and Log Monitoring System Using Raspberry Pi 4

Project Documentation

Prepared by

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1. Project Overview

This project involves building a lightweight Security Information and Event Management (SIEM) system using a Raspberry Pi 4 as the central server. The goal was to set up real-time log collection from both a Linux device (the Pi itself) and a Windows 10 laptop, store everything in a single place, and get email alerts whenever suspicious activity was detected.

The entire stack runs on local hardware with no cloud dependency. Grafana Loki is used for log storage, Grafana for dashboards and alerting, Fluent Bit for collecting Pi system logs, and Winlogbeat + Promtail for forwarding Windows Event logs to the Pi.

1.1 Device Setup

Device	Model / OS	Role
Raspberry Pi 4	Raspberry Pi OS (64-bit), 4GB RAM, 64GB SD	SIEM Server+ Monitored Device
Windows Laptop	Windows 10	Monitored Endpoint

1.2 Software Used

Tool	Version	What It Does
Grafana Loki	v2.9.0	Stores all logs from Pi and Windows
Grafana	Latest	Dashboard, alert rules, email notifications
Fluent Bit	v4.2.3	Reads Raspberry Pi system logs and sends to Loki
Winlogbeat	v9.3.1	Reads Windows Event Viewer and writes to .ndjson files
Promtail	v3.6.7	Reads Winlogbeat files and ships them to Loki
Docker Compose	V2	Runs Loki and Grafana as containers on the Pi
NSSM	v2.24	Installs Promtail as an auto-starting Windows service
Gmail SMTP	-	Sends email alerts from Grafana

2. Architecture

Log data flows from two sources into Loki on the Pi, where Grafana queries it and raises alerts.

From the Windows laptop: Winlogbeat reads Windows Event Viewer and writes structured JSON files to disk. Promtail watches those files and pushes new entries to Loki over HTTP.

From the Raspberry Pi itself: Fluent Bit reads the systemd journal (Pi OS does not use /var/log/auth.log — it stores everything in the journal) and forwards logs directly to Loki.

Grafana sits on top of Loki, running alert rules every minute. When a rule fires, Grafana sends an email via Gmail SMTP to the configured contact point.

```
Windows PC
  └── Winlogbeat   →  writes logs to C:\winlogbeat\logs\*.ndjson
  └── Promtail     →  ships .ndjson to Loki on Pi
          |
          ▼
Raspberry Pi  (SIEM Server)
  └── Fluent Bit   →  reads systemd journal → Loki
  └── Loki         →  stores all logs (Pi + Windows)
  └── Grafana      →  dashboard + 7 alert rules
       └── Gmail SMTP  →  Email alerts
```

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3. Installation Steps

3.1 Raspberry Pi — Docker & SIEM Stack

First update the system, then install Docker and start the containers:

```
sudo apt update && sudo apt upgrade -y
curl -fsSL https://get.docker.com | sh
sudo usermod -aG docker $USER
```

Create the project folder and docker-compose.yml:

```
mkdir ~/siem-lab && cd ~/siem-lab
nano docker-compose.yml
```

Paste the following into docker-compose.yml:

```
services:
  loki:
    image: grafana/loki:2.9.0
    container_name: loki
    restart: always
    ports:
      - "3100:3100"
    volumes:
      - loki-data:/loki

  grafana:
    image: grafana/grafana:latest
    container_name: grafana
    restart: always
    ports:
      - "3000:3000"
    volumes:
      - grafana-data:/var/lib/grafana
    environment:
      - GF_SMTP_ENABLED=true
      - GF_SMTP_HOST=smtp.gmail.com:587
      - GF_SMTP_USER=your_email@gmail.com
      - GF_SMTP_PASSWORD=your_app_password
      - GF_SMTP_FROM_ADDRESS=your_email@gmail.com
      - GF_SMTP_FROM_NAME=SIEM Alert

volumes:
  grafana-data:
  loki-data:
```

```
docker compose up -d
```

Important: Use 'docker compose' (V2) not 'docker-compose' (V1). The volumes section is critical — without it, all Grafana settings are lost on restart.

3.2 Raspberry Pi — Fluent Bit

Fluent Bit collects system logs from the Pi and forwards them to Loki. On Raspberry Pi OS, logs are stored in the systemd journal — not in /var/log/auth.log. The config must use the systemd input type.

```
sudo apt install fluent-bit -y
sudo nano /etc/fluent-bit/fluent-bit.conf
```

Replace the contents with:

```
[SERVICE]
    Flush        5
    Daemon       Off
    Log_Level    info

[INPUT]
    Name           systemd
    Tag            syslog
    Systemd_Filter _TRANSPORT=syslog
    Read_From_Tail On

[INPUT]
    Name           systemd
    Tag            auth
    Systemd_Filter _COMM=sshd
    Read_From_Tail On

[OUTPUT]
    Name   loki
    Match  *
    Host   127.0.0.1
    Port   3100
    Labels job=fluentbit
```

```
sudo systemctl enable fluent-bit
sudo systemctl start fluent-bit
```

3.3 Windows — Winlogbeat

Winlogbeat reads the Windows Event Viewer and writes logs as JSON files. Note: Winlogbeat v9.x does not support HTTP output, so the output is set to file and Promtail handles delivery to Loki.

- Download Winlogbeat v9.3.1 from elastic.co and extract to C:\winlogbeat\
- Edit C:\winlogbeat\winlogbeat.yml — set output to file

Key section of winlogbeat.yml:

```
output.file:
  path: C:\winlogbeat\logs
  filename: winlogbeat-output
  rotate_every_kb: 10000
  number_of_files: 7
```

Install as a Windows service (run PowerShell as Administrator):

```
cd C:\winlogbeat
.\install-service-winlogbeat.ps1
Set-Service -Name winlogbeat -StartupType Automatic
Start-Service winlogbeat
```

3.4 Windows — Promtail

Promtail watches the Winlogbeat output files and ships new entries to Loki. It is installed as a Windows service using NSSM to ensure it starts automatically on boot.

- Download Promtail v3.6.7 (promtail-windows-amd64.exe) from Grafana GitHub releases — extract to C:\promtail\
- Download NSSM v2.24 from nssm.cc — copy nssm.exe (from the win64 folder) to C:\promtail\
- Create C:\promtail\promtail-config.yml with the following:

```
server:
  http_listen_port: 9081
  grpc_listen_port: 9096

positions:
  filename: C:\promtail\positions.yaml

clients:
  - url: http://192.168.0.6:3100/loki/api/v1/push

scrape_configs:
  - job_name: windows
    static_configs:
      - targets:
          - localhost
        labels:
          job: windows-logs
          host: LAPTOP-D48G2U41
          __path__: C:\winlogbeat\logs\*.ndjson
```

Install Promtail as a Windows service:

```
cd C:\promtail
.\nssm.exe install Promtail-SIEM "C:\promtail\promtail-windows-amd64.exe"
"--config.file=C:\promtail\promtail-config.yml"
.\nssm.exe set Promtail-SIEM Start SERVICE_AUTO_START
Start-Service Promtail-SIEM
```

4. Grafana Setup

4.1 Adding Loki as a Datasource

Open Grafana at http://192.168.0.6:3000. Default credentials are admin / admin. Navigate to Connections → Data Sources → Add new datasource → select Loki.

For the URL, enter http://loki:3100 — not localhost. Since Grafana runs inside a Docker container, 'localhost' would refer to the container itself, not the Loki container. Docker Compose makes containers reachable by their service names.

4.2 Gmail SMTP for Email Alerts

Email alerting is configured through Docker environment variables in docker-compose.yml. A Gmail App Password is required — the regular Gmail password will not work. To generate one: Google Account → Security → 2-Step Verification → App Passwords.

After updating the password in docker-compose.yml, restart Grafana:

```
cd ~/siem-lab && docker compose up -d --force-recreate grafana
```

In Grafana, go to Alerting → Contact points → Add contact point. Name it SIEM-Alerts, set type to Email, and enter your address. Send a test email to confirm it works.

4.3 Dashboard Panels

The SOC Security Operations Dashboard contains 10 panels. The four stat counters at the top give a quick summary; the two time-series graphs show attack patterns over time; and the log viewers at the bottom show raw events.

Panel Name	Type	LogQL Query
Total Linux (Pi) Logs	Stat	{job="fluentbit"}[24h]
Windows Events (24h)	Stat	{job="windows-logs"}[24h]
Failed SSH Logins (24h)	Stat	{job="fluentbit"} |= "Failed password" [24h]
Sudo Commands (24h)	Stat	{job="fluentbit"} |= "sudo" [24h]
SSH Brute Force Timeline	Time series	{job="fluentbit"} |= "Failed password" [5m]
Windows Login Failures	Time series	{job="windows-logs"} |= "failure" [5m]
Live Raspberry Pi Logs	Logs	{job="fluentbit"}
Live Windows Event Logs	Logs	{job="windows-logs"}
SSH Attack Logs	Logs	{job="fluentbit"} |= "Failed password"
Sudo Command Logs	Logs	{job="fluentbit"} |= "sudo"

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5. Detection Rules

Seven alert rules were created in Grafana under the SIEM folder. Each rule uses a LogQL query to count matching log entries over a 5-minute window. Rules are evaluated every minute and send an email via the SIEM-Alerts contact point when they fire.

Rule Name	Log Source	Threshold	Folder Group
SSH Brute Force Detection	Pi (fluentbit)	More than 5 in 5 min	linux-auth
Sudo Abuse Detection	Pi (fluentbit)	More than 10 in 5 min	linux-auth
New User Created (Pi)	Pi (fluentbit)	Any occurrence	linux-auth
System Reboot Detected	Pi (fluentbit)	Any occurrence	linux-system
Windows Brute Force	Windows (windows-logs)	More than 5 in 5 min	windows-auth
New User Created (Windows)	Windows (windows-logs)	Any occurrence	windows-auth
Windows Privilege Escalation	Windows (windows-logs)	Any occurrence	windows-auth

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6. MITRE ATT&CK Mapping

Each detection rule is mapped to a MITRE ATT&CK technique. This gives the rules a standard reference that any SOC team would recognise and makes the alerts more actionable during investigation.

Rule	MITRE ID	Technique	Tactic
SSH Brute Force	T1110	Brute Force	Credential Access
Sudo Abuse	T1548	Abuse Elevation Control	Privilege Escalation
New User (Pi)	T1136.001	Create Local Account	Persistence
System Reboot	T1529	System Shutdown / Reboot	Impact
Windows Brute Force	T1110.001	Password Guessing	Credential Access
New User (Windows)	T1136.001	Create Local Account	Persistence
Windows Privilege Esc	T1134	Access Token Manipulation	Privilege Escalation

The rules cover four MITRE tactics: Credential Access, Privilege Escalation, Persistence, and Impact. Together they span the key stages of an attack — from initial access attempts through to maintaining a foothold and causing disruption.

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7. Screenshots

7.1 SOC Security Operations Dashboard

The main dashboard at the time of capture. The Pi had generated 414 log entries in the last 24 hours, with 11 failed SSH login attempts and 9 sudo commands recorded. The Windows endpoint had forwarded over 118,000 events.

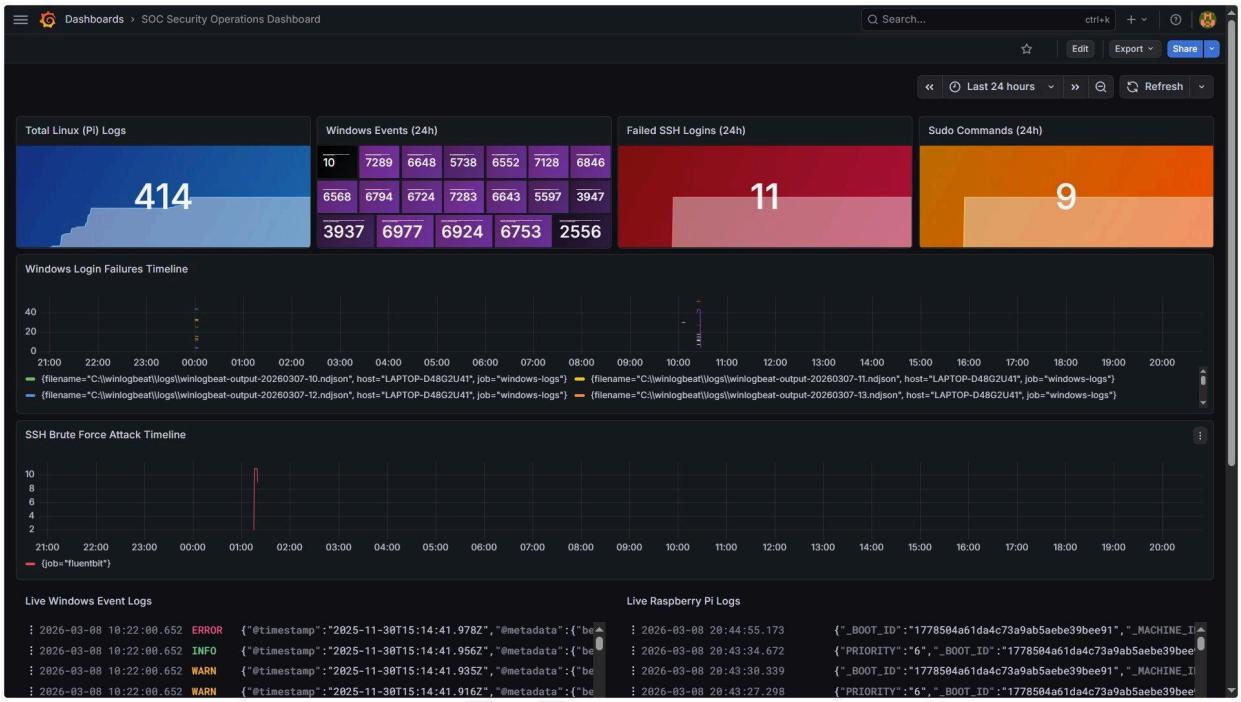

Figure 1 — SOC Security Operations Dashboard showing real-time data from both devices

7.2 Alert Rule — SSH Brute Force Detection

The alert rule editor showing the LogQL query, alert condition, MITRE ATT&CK annotation, and notification contact point configured for the SSH Brute Force detection rule.

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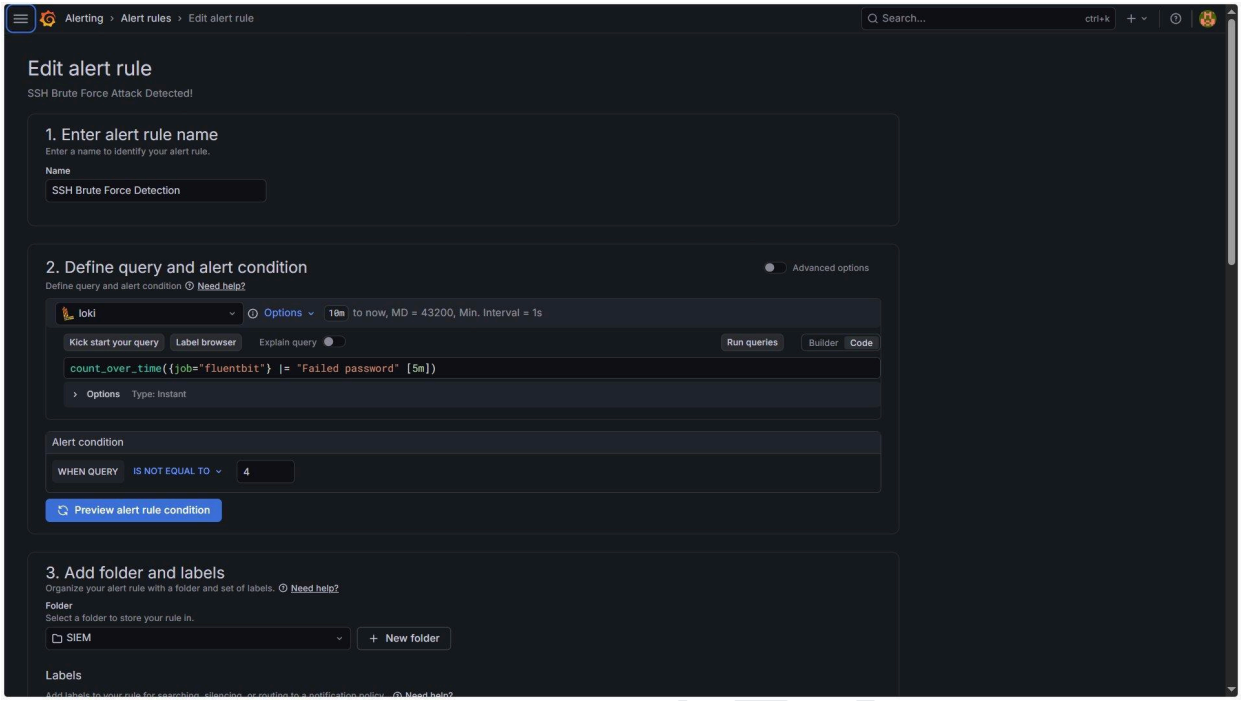

Figure 2 — Alert rule query and condition configuration

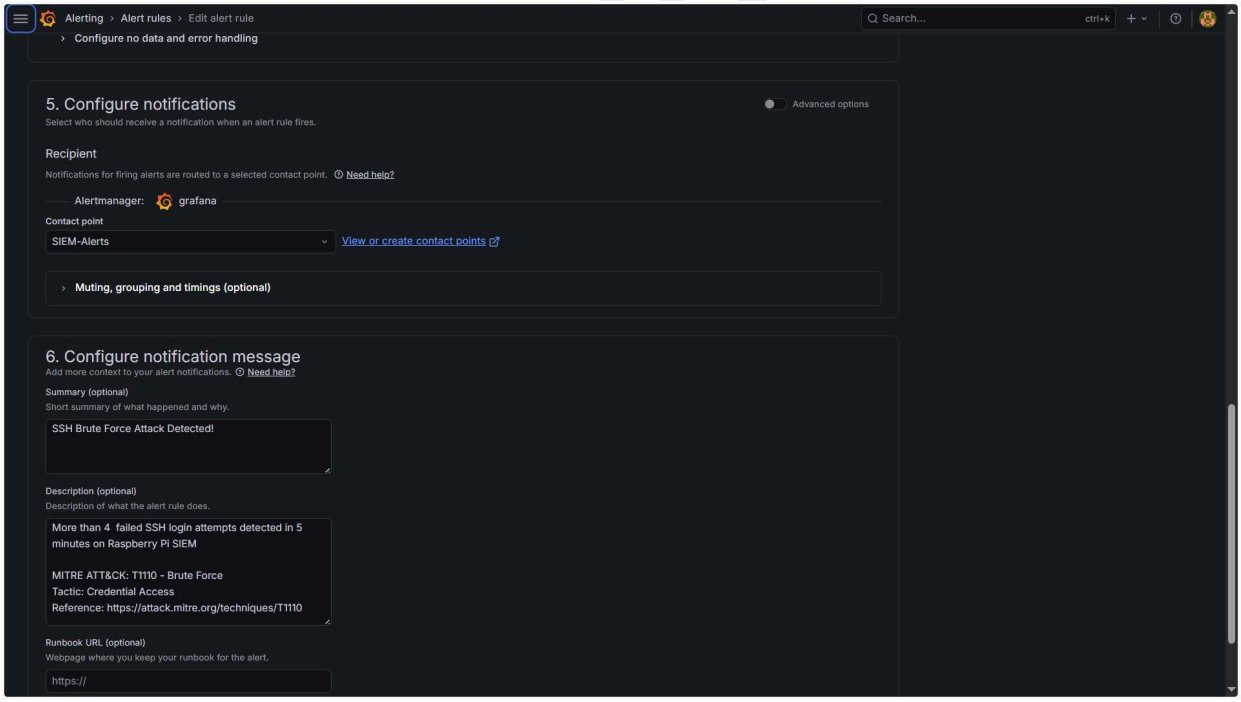

Figure 3 — Alert notification message with MITRE ATT&CK reference and contact point

7.3 Email Alert

An example email received when the Windows Privilege Escalation rule fired. The email is sent by Grafana via Gmail SMTP and includes the alert summary, description, MITRE technique, and a direct link to view the alert in Grafana.

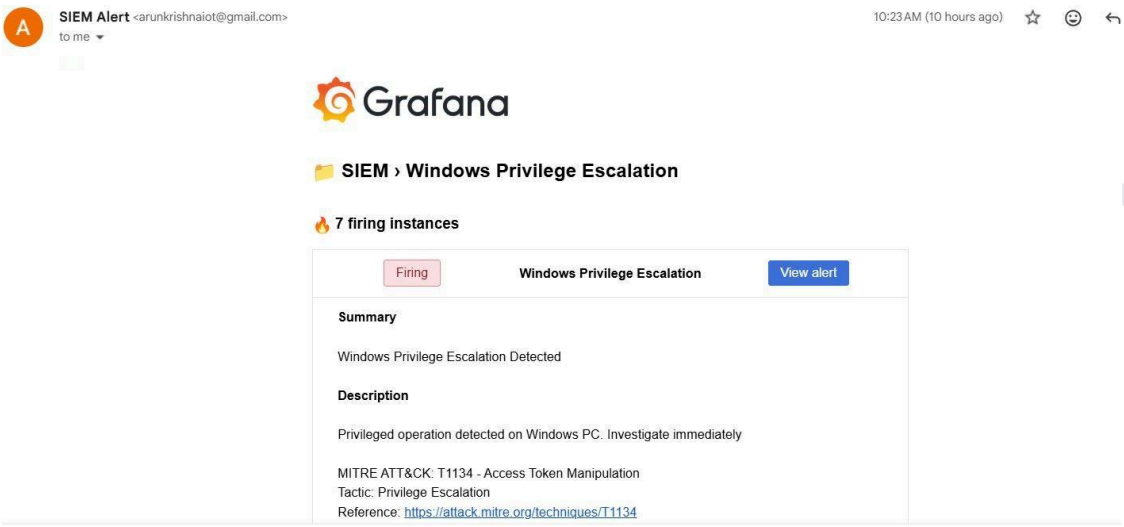

Figure 4 — Grafana email alert for Windows Privilege Escalation (T1134)

7.4 Log Exploration in Grafana

Raspberry Pi system logs being queried live in Grafana Explore. The query {job="fluentbit"} returned 414 structured log lines sourced from the Pi's systemd journal.

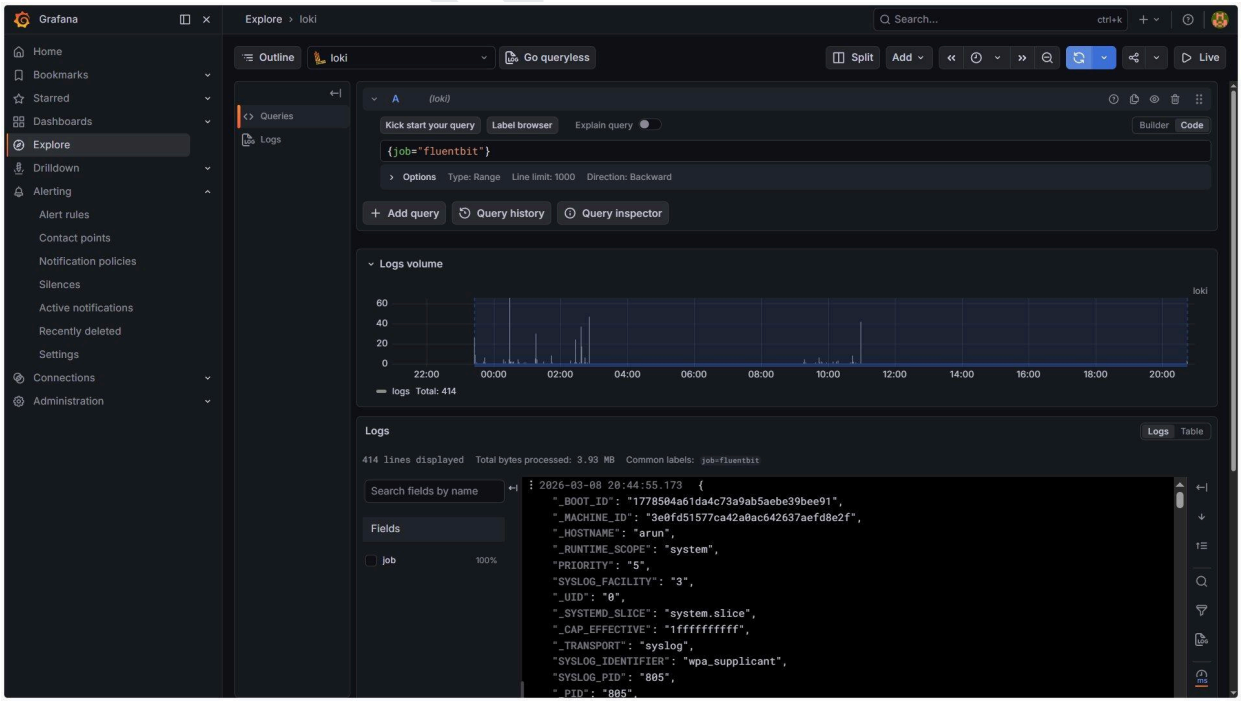

Figure 5 — Live Raspberry Pi logs in Grafana Explore

Windows Event logs from Winlogbeat via Promtail. The label job=windows-logs was used to separate Windows events from Pi logs within Loki.

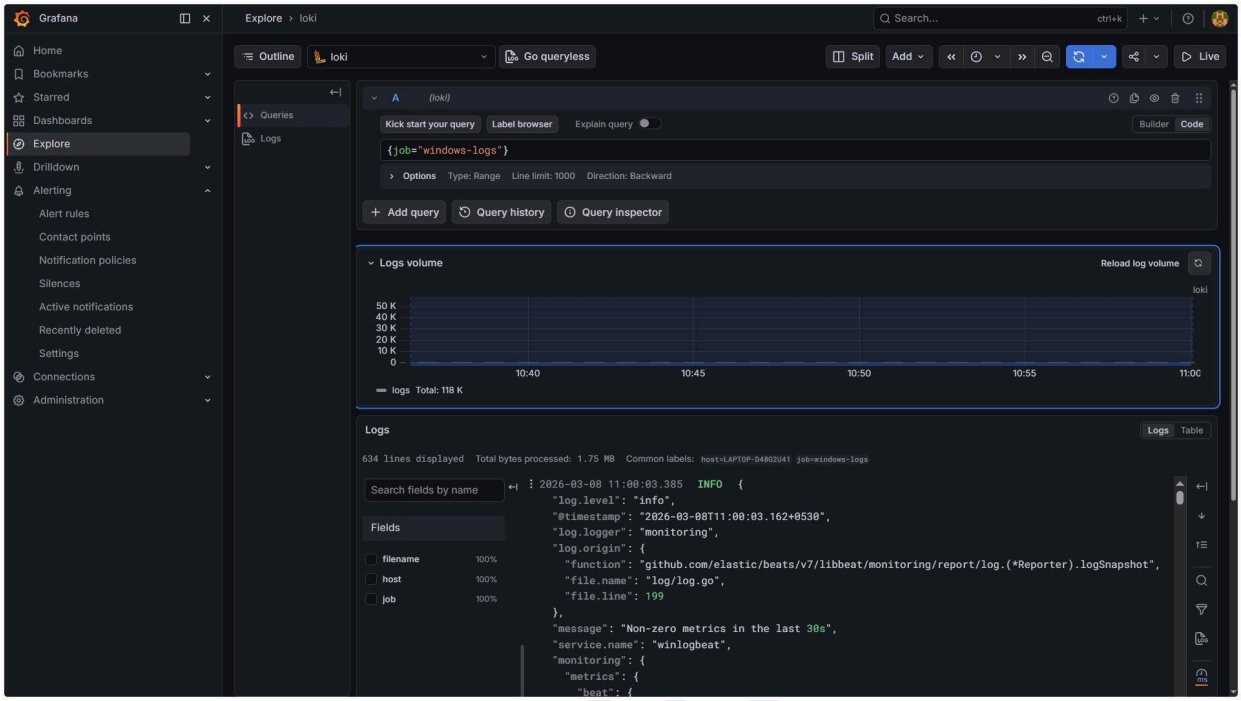

Figure 6 — Live Windows Event logs in Grafana Explore

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8. References

- Grafana Loki Documentation — https://grafana.com/docs/loki/latest/
- Fluent Bit Documentation — https://docs.fluentbit.io/manual/
- Winlogbeat Documentation — https://www.elastic.co/beats/winlogbeat
- Promtail Documentation — https://grafana.com/docs/loki/latest/send-data/promtail/
- MITRE ATT&CK Framework — https://attack.mitre.org/
- Docker Documentation — https://docs.docker.com/

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